# CONCEPT 1 — TERRACOTTA COURTYARD HOUSE

Two storeys in the ONT band between 3-storey neighbours · 1.5 m west service passage · east light court · shaded north balcony



Blender render — street view (golden hour)

A two-storey family house in northern Vietnamese timber, brick and terracotta, kept wholly inside the 13.5 × 19.7 m road-side ONT band. The entrance hall (Huyền Quan) is on the road. Living space runs north towards the garden. A 2.5 × 4 m east light court brings daylight and air into the middle of the deep plan. A 1.5 m service passage on the west side serves the wet kitchen and laundry without crossing the living rooms. Upstairs, three bedrooms, an altar room and a family lounge open onto a 2.2 m shaded balcony that looks over the existing garden land towards the river. It is designed for a Vietnamese street, with a 3-storey house assumed on each side. The east wall is blank because it faces the neighbour's party wall 0.6 m away. The west wall has only high-level windows onto the service passage. Daylight comes from the street, the light court and the garden. The only structure on the DNN land is the pair of boundary walls running down to the water.




Concept 1 brief image (inspiration)

## THE CONCEPT IN NUMBERS

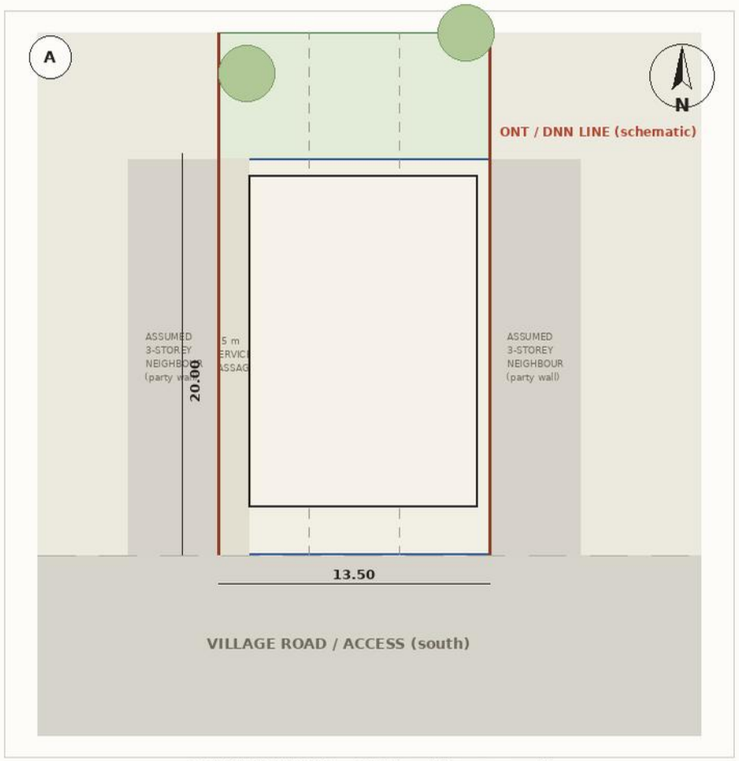
| | |
|---|---|
| **Footprint** | 188.1 m² (71% of ONT band) |
| **Ground floor** | 178 m² enclosed |
| **Upper floor** | 153 m² enclosed + 25 m² balcony |
| **Total enclosed** | 331 m² |
| **Constraint check** | PASS — all structures in ONT; DNN planting only |

**Walkthrough test: 24/24 rooms reachable**

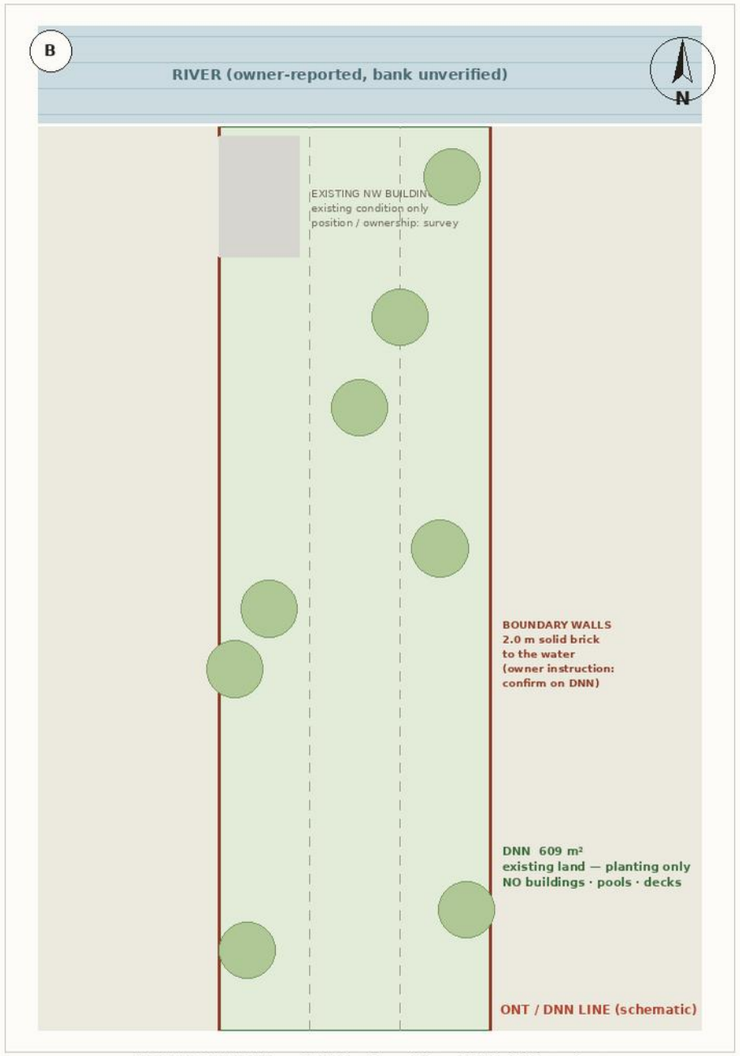
The concept image asked for about 230–250 m² on the ground floor. The ONT band is only about 266 m², so the plan is fitted inside it with a 1.5 m west service passage, a 2.4 m forecourt and 0.6 m eaves. That gives 178 m² enclosed on the ground floor. The house sits on the east boundary, which needs a boundary/setback check.

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

**SITE PLAN · 1:200 at A3 · shown in two parts so the narrow 13.5 m plot stays readable**




**SOUTHERLY PART — ONT band, house, road**




**NORTHERLY PART — DNN land, existing NW building, river**

**NOTES**

Schematic site from owner information only: three side-by-side 4.5 m titles, about 13.5 × 65 m. No survey. Corners, levels, bank edge and road reserve are unverified.

Structures are shown ONLY inside the ONT road-side band (3 × 88.8 = 266.4 m²). The 609 m² DNN band is existing land, shown as planting only.

The title sketch shows 6.5 m to the road centreline. It is NOT treated as a verified setback.

Assumed: a 3-storey house on each side built to the property line to about the same 20 m depth. So the east wall is blank (party wall) and the west wall has high-level windows only.

Side boundary walls run the full length to the water on both property lines (owner instruction). This is the only structure on DNN land. Confirm it is permitted.

The existing north-west building is an existing condition only. Its position and ownership are to be surveyed. It is not altered.

Subject to legal merger of the three ONT sections, setbacks, river corridor and permits.

**KEY**

- ONT — buildable band
- DNN — land / planting only
- service passage
- existing NW building
- assumed 3-storey neighbours
- boundary walls
- house footprint
- assumed / context

0 10 20 30 40 50 metres (assumed — no survey)

## CONCEPT 1 — TERRACOTTA COURTYARD HOUSE

Two storeys in the ONT band between 3-storey neighbours · 1.5 m west service passage · east light court · shaded north bale

| SCALE | SHEET | STATUS |
|---|---|---|
| **1:200 at A3** | **C1-01** site plan | **FOR REVIEW** *not for construction* |

enclosed 178.1 m² · level 1 of 2

## GROUND FLOOR · FFL +0.45 / +3.75




enclosed 153.0 m² · level 2 of 2

## UPPER FLOOR · FFL +3.75 / +6.85




0 10 20 30 40 50 metres (assumed — no survey)

### KEY

- HABITABLE
- SERVICE / WET
- CIRCULATION
- STORAGE
- SPECIAL
- OUTDOOR
- PLANT / VOID

### NOTES

CONCEPT ONLY — NOT FOR CONSTRUCTION. Dimensions are taken from owner-supplied title information; no survey. Subject to survey, legal merger, setbacks and permits.

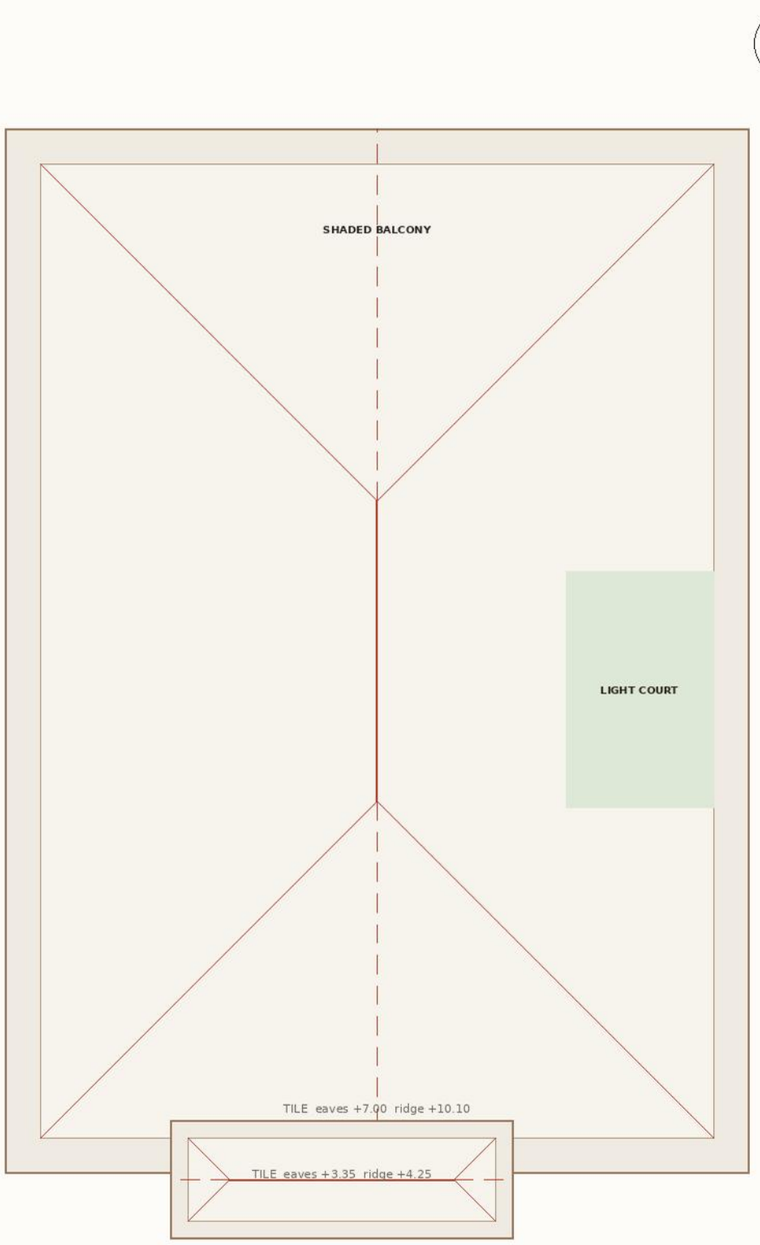
Areas are measured from the room rectangles; the balcony and light court are excluded from enclosed area.

## CONCEPT 1 — TERRACOTTA COURTYARD HOUSE

Two storeys in the ONT band between 3-storey neighbours · 1.5 m west service passage · east light court · shaded north balc

| SCALE | SHEET | STATUS |
|---|---|---|
| 1:100 at A3 | C1-02 | FOR REVIEW |
| | floor plans 1–2 | not for construction |

## ROOF PLAN · looking down




0 10 20 30 40 50 metres (assumed — no survey)

**KEY**

- roof covering
- ridge / hip
- fascia / eaves
- open slot below

**ROOF KEY**

- concrete tile on timber — eaves +7.00, ridge +10.10
- ridge / hip line
- fascia / eaves edge, 1.05 m overhang
- rain-garden slot, open to the sky
- outline of the level below (hidden line)

**NOTES**

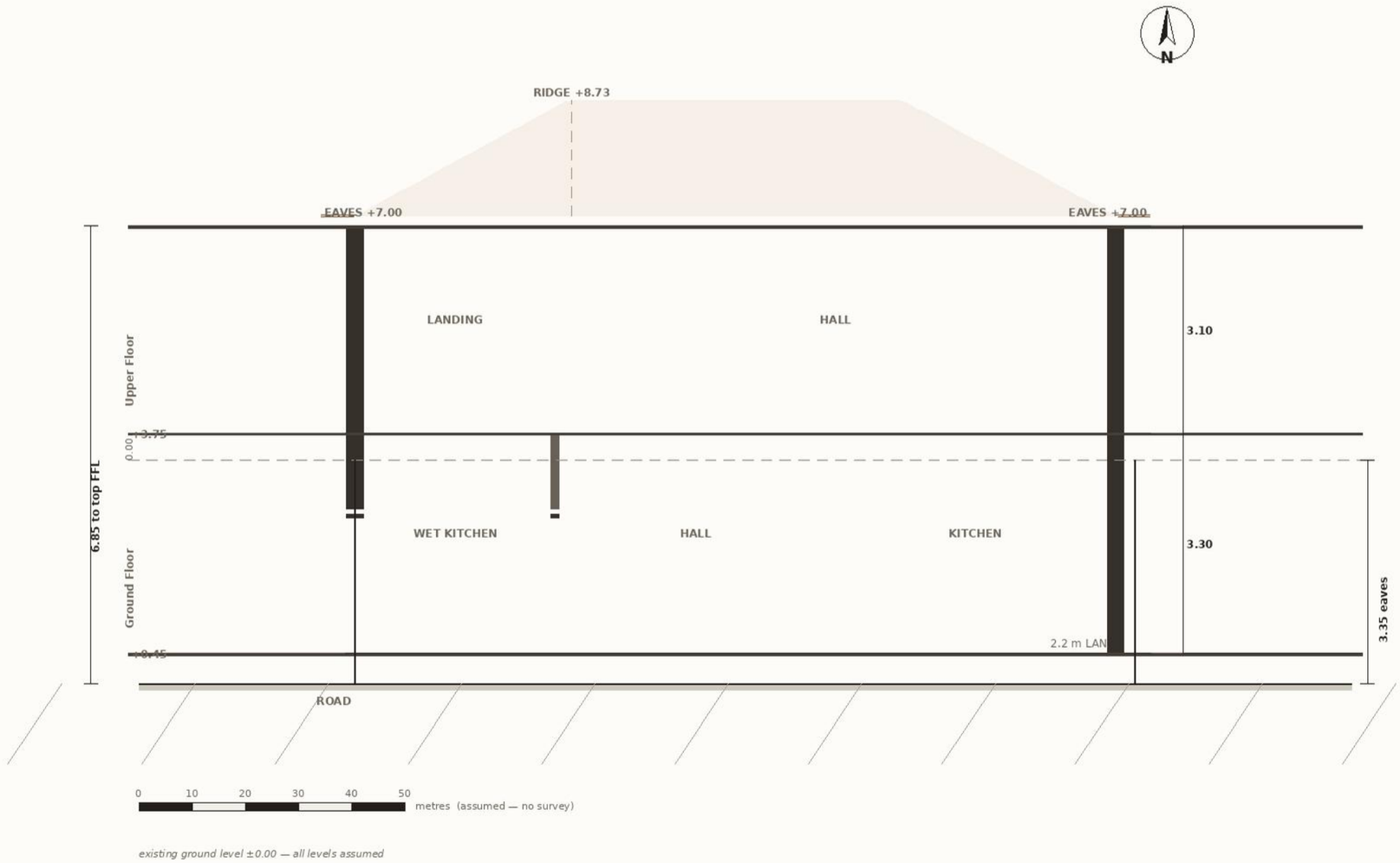
Ridge and eaves heights come from the option massing: eaves +7.00 m, ridge +10.10 m above datum, 0.60 m overhang all round.
Hips, ridges and the rain-garden slot are derived from the same rectangles as the plans and the 3D model.
Rainwater from the slot and the roof gardens is the only visible drainage; confirm the neighbour-boundary falls before detailing.

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|---|---|---|
| **1:100 at A3** | **C1-03**<br>roof plan | **FOR REVIEW**<br>*not for construction* |

## SECTION A–A · looking north at y = 7.00 m (assumed cut)




*existing ground level ±0.00 — all levels assumed*

**KEY**

- cut external wall, 0.30
- cut internal wall, 0.15
- glazing in the cut
- roof build-up in section

**NOTES**

Datum 0.00 = existing road and ground level. Finished ground floor +0.45 m, first +3.75 m, eaves +3.35 m, ridge +10.10 m.

Slabs, walls and levels come from the same rectangles as the plans, so the section and the plans can never disagree.

The 2.2 m rain-garden slot is open to the sky on every level: draw it as a full-height void when the structure is detailed.

The cut is at y = 7.00 m. Move it on the plans to suit a later survey.

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|---|---|---|
| **1:100 at A3** | **C1-04**<br>section A–A | **FOR REVIEW**<br>*not for construction* |

## AREA SCHEDULE · net internal, measured from the plan rectangles

| Room | Type | Area m² | Cum. m² |
|---|---|---|---|
| **Ground Floor · FFL +0.45** | | | |
| Dining | Habitable | 23.6 | 23.6 |
| Robe | Habitable | 5.8 | 29.4 |
| Living (garden view) | Habitable | 33.7 | 63.0 |
| Master Suite | Habitable | 21.6 | 84.6 |
| Light Court | Plant / Void | 10.0 | 94.6 |
| Ensuite | Service / Wet | 5.8 | 100.4 |
| Gallery | Circulation | 11.9 | 112.3 |
| Entry (Huyền Quan) | Circulation | 10.1 | 122.3 |
| Hall | Circulation | 13.4 | 135.8 |
| Kitchen | Service / Wet | 13.4 | 149.2 |
| Laundry / WC | Service / Wet | 7.2 | 156.4 |
| Pantry / Store | Storage | 10.1 | 166.5 |
| Stair | Circulation | 12.0 | 178.5 |
| Wet Kitchen | Service / Wet | 9.6 | 188.1 |
| Level total, enclosed | | **178.1** | |
| **Upper Floor · FFL +3.75** | | | |
| Bedroom 1 | Habitable | 16.0 | 16.0 |
| Bedroom 2 | Habitable | 16.0 | 31.9 |
| Bedroom 3 | Habitable | 19.7 | 51.7 |
| Family Lounge | Habitable | 19.7 | 71.4 |
| Altar Room | Special | 14.1 | 85.5 |
| Shaded Balcony | Outdoor | 25.1 | 110.6 |
| Bathroom | Service / Wet | 7.2 | 117.8 |
| Bathroom 2 | Service / Wet | 7.8 | 125.6 |
| Light Court | Plant / Void | 10.0 | 135.6 |
| Hall | Circulation | 15.1 | 150.7 |
| Gallery | Circulation | 11.6 | 162.3 |
| Landing | Circulation | 9.6 | 171.9 |
| Linen | Storage | 4.2 | 176.1 |
| Stair | Circulation | 12.0 | 188.1 |
| Level total, enclosed | | **153.0** | |

| AREA SUMMARY | Area |
|---|---|
| Level 1 — Ground Floor (enclosed) | 178.1 m² |
| Level 2 — Upper Floor (enclosed) | 153.0 m² |
| **Total enclosed floor area** | **331.1 m²** |

### ORDER OF COST — TURNKEY, NORTHERN VIET NAM

| INDICATIVE BUILD COST | M VND | +10% prov. |
|---|---|---|
| Basic build — 4.9 M VND/m² | 1 622 | 1 785 |
| Mid build — 6.5 M VND/m² | 2 152 | 2 368 |
| High build — 8.5 M VND/m² | 2 815 | 3 096 |

**NOTES**

Areas are net internal floor areas computed from the room rectangles. Terraces, loggias, the rain-garden slot, the plant deck and roof gardens are excluded from the totals but shown on the plans.

The build-cost range is a rule of thumb from published Vietnamese cost guides for northern Viet Nam, not a quotation. It excludes site works, boundary treatment and furniture.

Confirm with a local quantity surveyor and at least two builders before committing to a budget.

**NOTES**

No habitable room here is below 2.7 m clear. The ground floor is a complete one-level home — bedroom, bathroom, living, dining and kitchen all at +0.45 — so the family can live on one level for as long as they want to.

Order-of-cost at 4.9–8.5 M VND/m² turnkey (northern Viet Nam) is in the table on the left. It is a rule of thumb, not a quotation.

**ROOM TYPE KEY**

- HABITABLE
- SERVICE / WET
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- STORAGE
- SPECIAL
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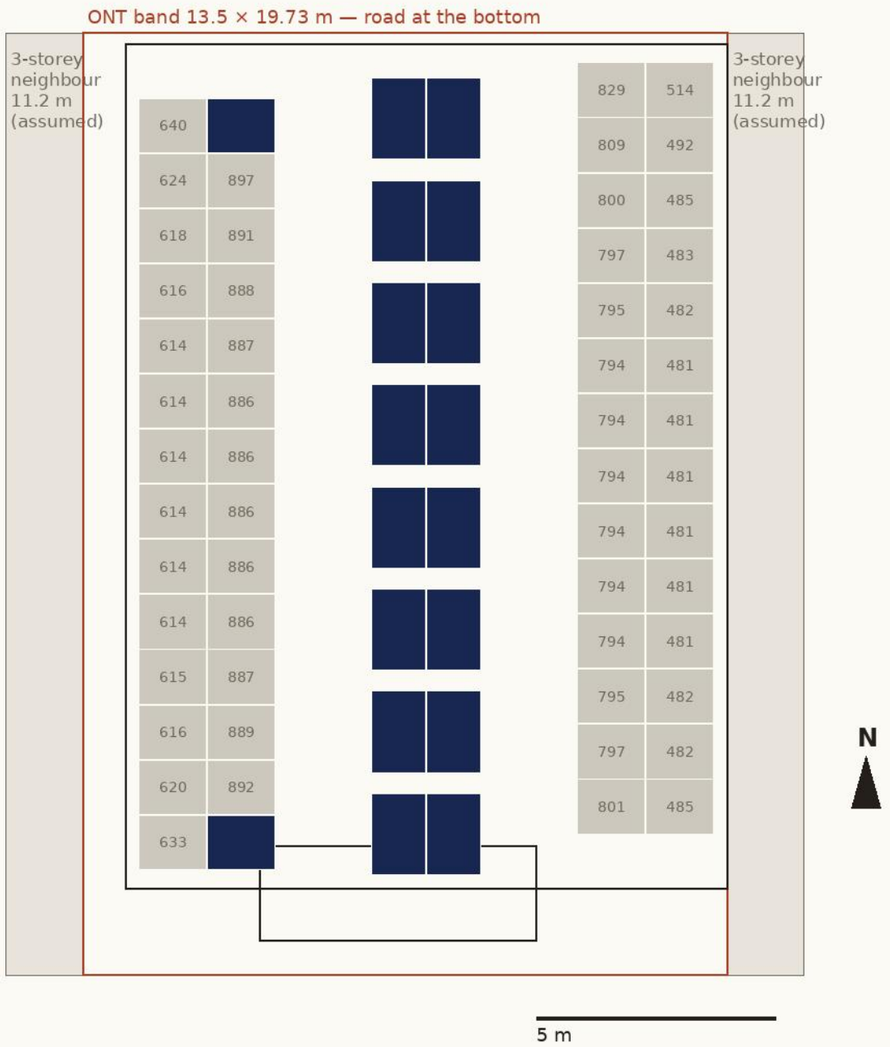
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|---|---|---|
| **as noted** | **C1-05**<br>area schedule | **FOR REVIEW**<br>*not for construction* |

# Rooftop solar — energy per season

18 × 440 Wp panels = 7.9 kWp on the roof · latitude 21°N (rounded) · PVGIS 5.3 TMY (ERA5), typical year · shaded by the assumed 3-storey neighbours




Recommended panels

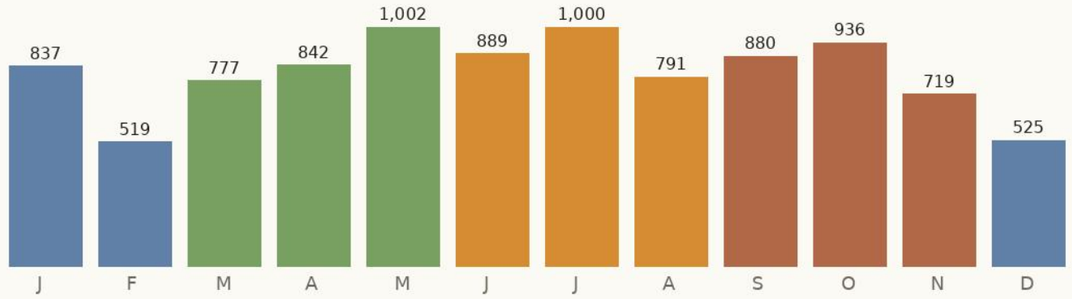
Fit but not worth it: number = kWh per kWp a year (an unshaded ideal panel gives 1281)

| Roof surface | Tilt / faces | Fit / used | kWh/kWp | no neighbours |
|---|---|---|---|---|
| Main roof west slope | 35° W | 28 / 2 | 755 | 1134 |
| Main roof flat top | 10° S | 16 / 16 | 1267 | 1270 |
| Main roof east slope | 35° E | 28 / 0 | 642 | 1089 |

## ENERGY FROM THE RECOMMENDED ARRAY

| Season | Months | kWh | kWh / day | Share |
|---|---|---|---|---|
| Winter | Dec–Feb | 1,881 | 20.9 | 19 % |
| Spring | Mar–May | 2,621 | 28.5 | 27 % |
| Summer | Jun–Aug | 2,680 | 29.1 | 28 % |
| Autumn | Sep–Nov | 2,535 | 27.9 | 26 % |
| **Year** | | **9,718** | **26.6** | |

Without the assumed neighbours the same panels would give 9,939 kWh a year (+2 %). All 72 panels that physically fit would give 26,129 kWh, but the extra ones are on shaded or east/west/north slopes.




## IS IT WORTH IT? (indicative)

| | Recommended | Best-value size |
|---|---|---|
| Panels / kWp | 18 / 7.9 | 7 / 3.1 |
| Installed cost | 111 M VND | 43 M VND |
| Bill saving, first year | 9 M VND | 8 M VND |
| Output used in the house | 26 % | 54 % |
| Share of the bill-kWh covered | 36 % | 31 % |
| Simple payback | 14 years | 6 years |
| Net over 25 years | 76 M VND | 128 M VND |

Assumes: an example air-conditioned family using about 6,890 kWh a year (bill 21 M VND a year), EVN household tiers from 1 Nov 2026 + 8 % VAT, 14 M VND per kWp installed, nothing paid for power sent to the grid (the buy-back rate is not yet published), 0.5 %/yr panel ageing, 1 %/yr upkeep. Surplus power is the limit: the house uses most of its power in the evening. Try other numbers at ali3b.github.io/home-design/solar.html.

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# Blender views

Cycles renders from the same data as the drawings and the walkthrough



Street view



From the garden (DNN land, planting only)

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# Blender views

Cycles renders from the same data as the drawings and the walkthrough



From the river end, between the boundary walls

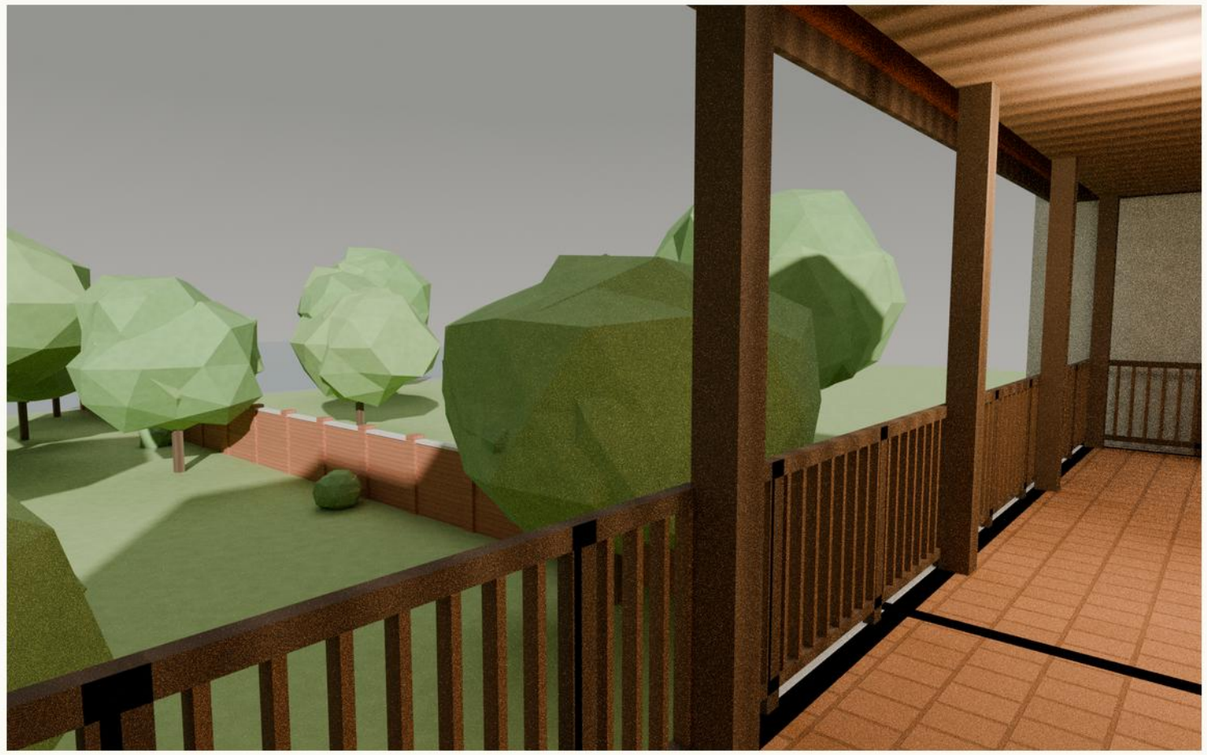


Aerial

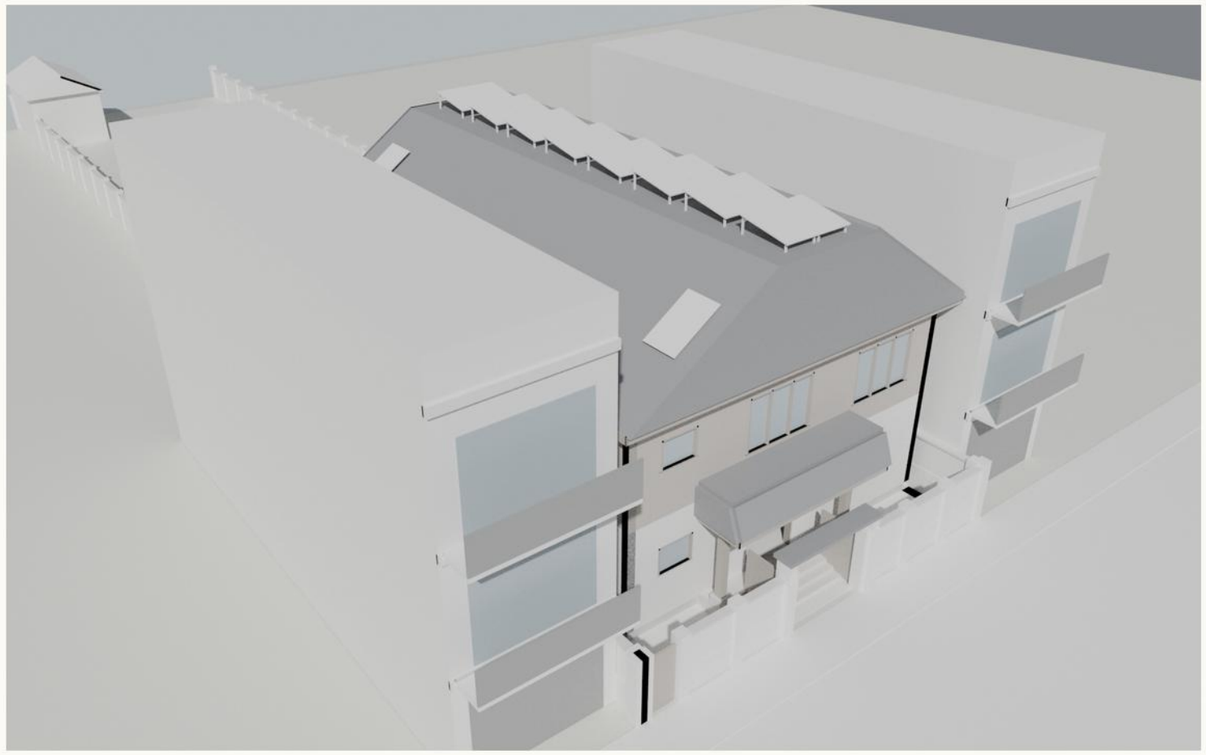
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# Blender views

Cycles renders from the same data as the drawings and the walkthrough



Upper balcony looking north



Clay massing

# Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



Street



Entry



Living



Kitchen

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# Walkthrough (Unity)

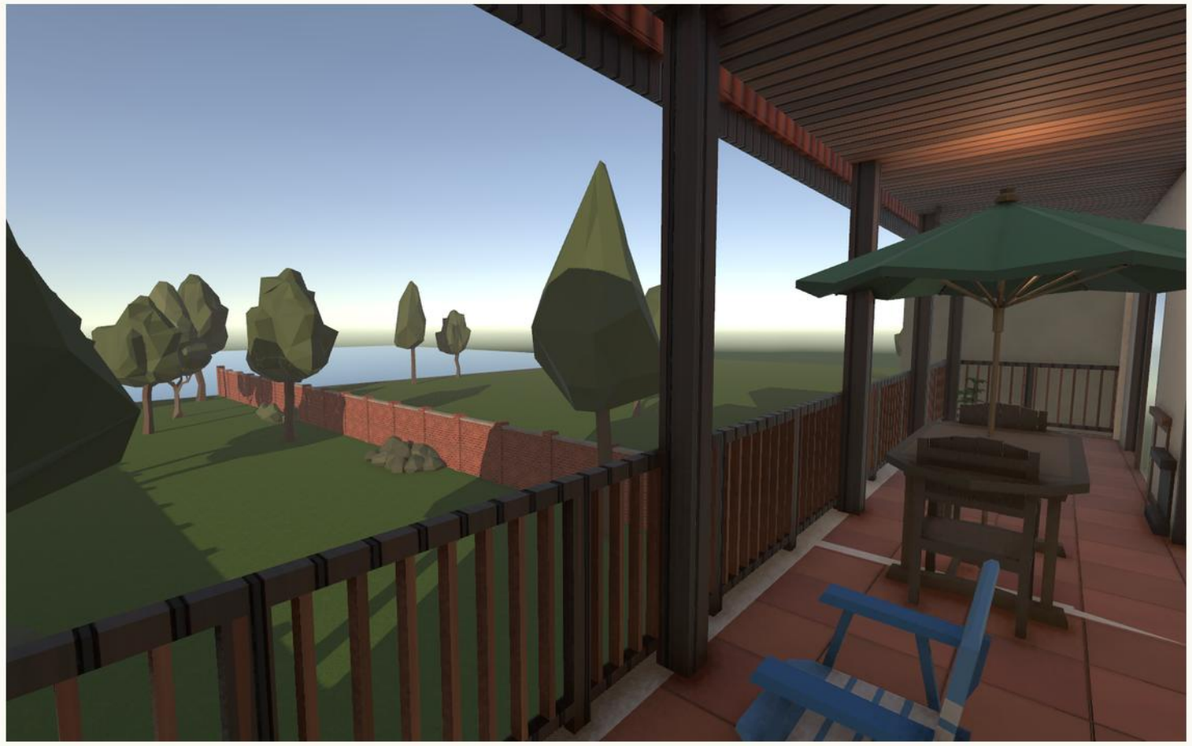
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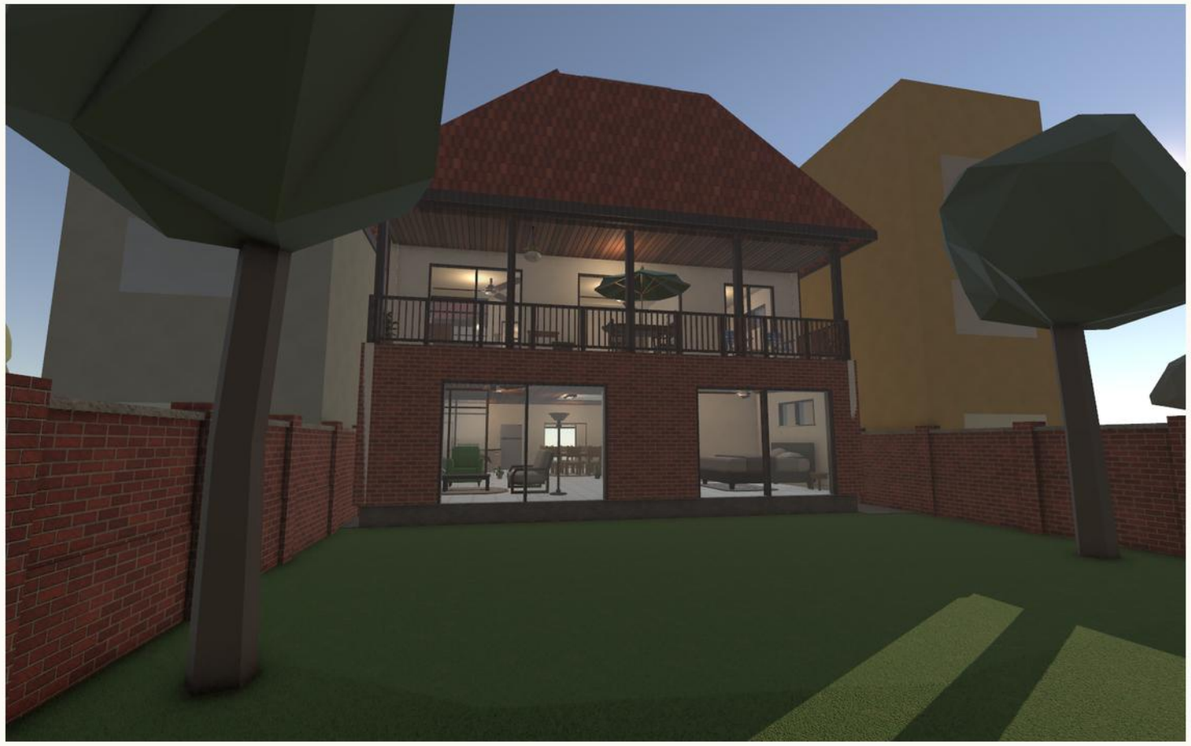
Master



Lounge



Balcony



Garden

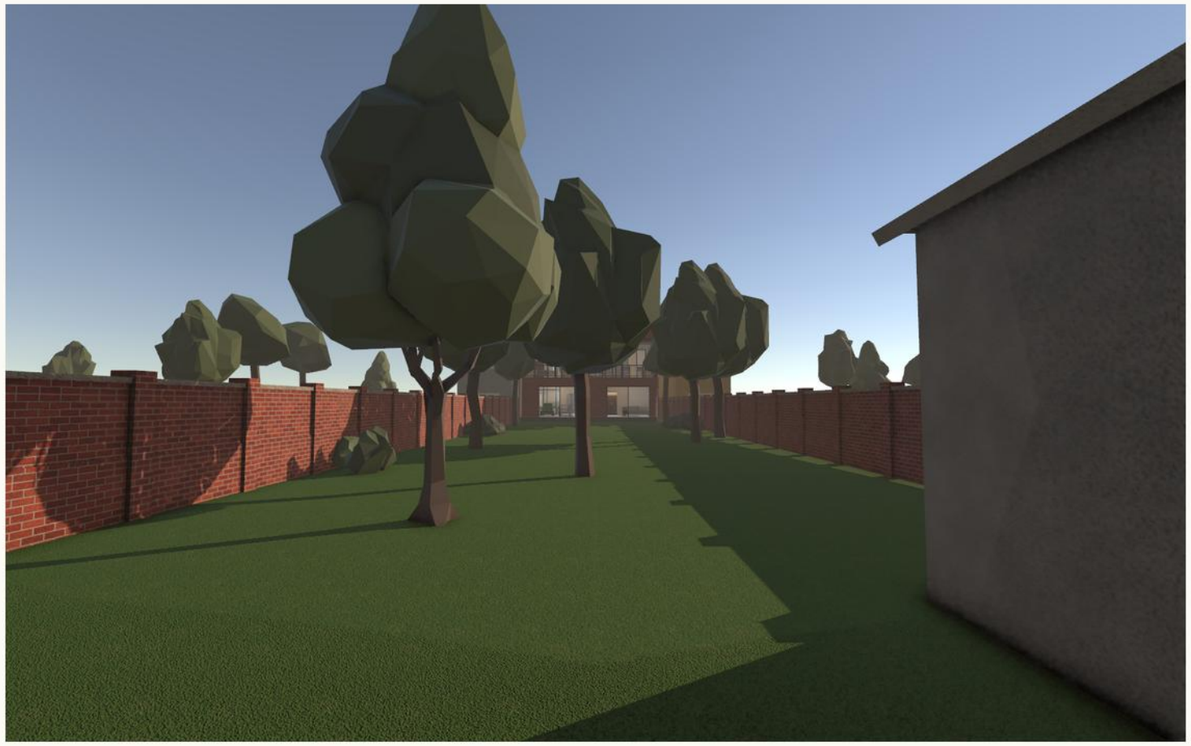
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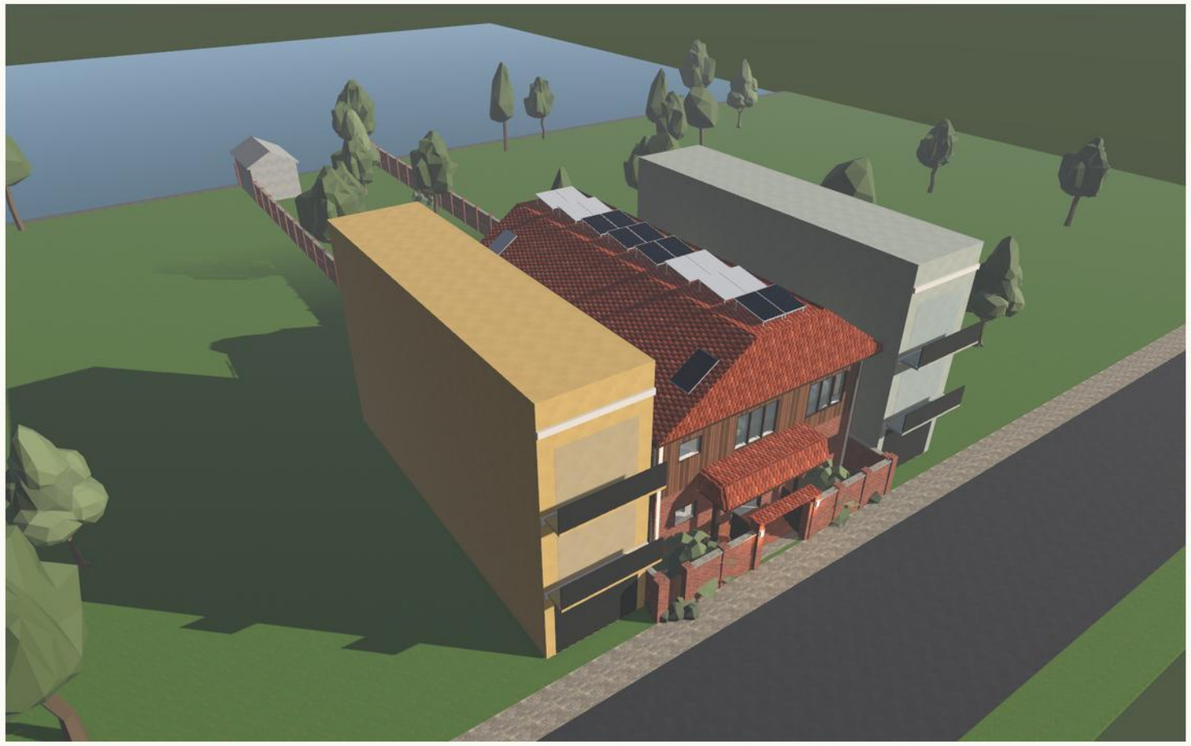
Aerial



Riverside



Dollhouse Ground

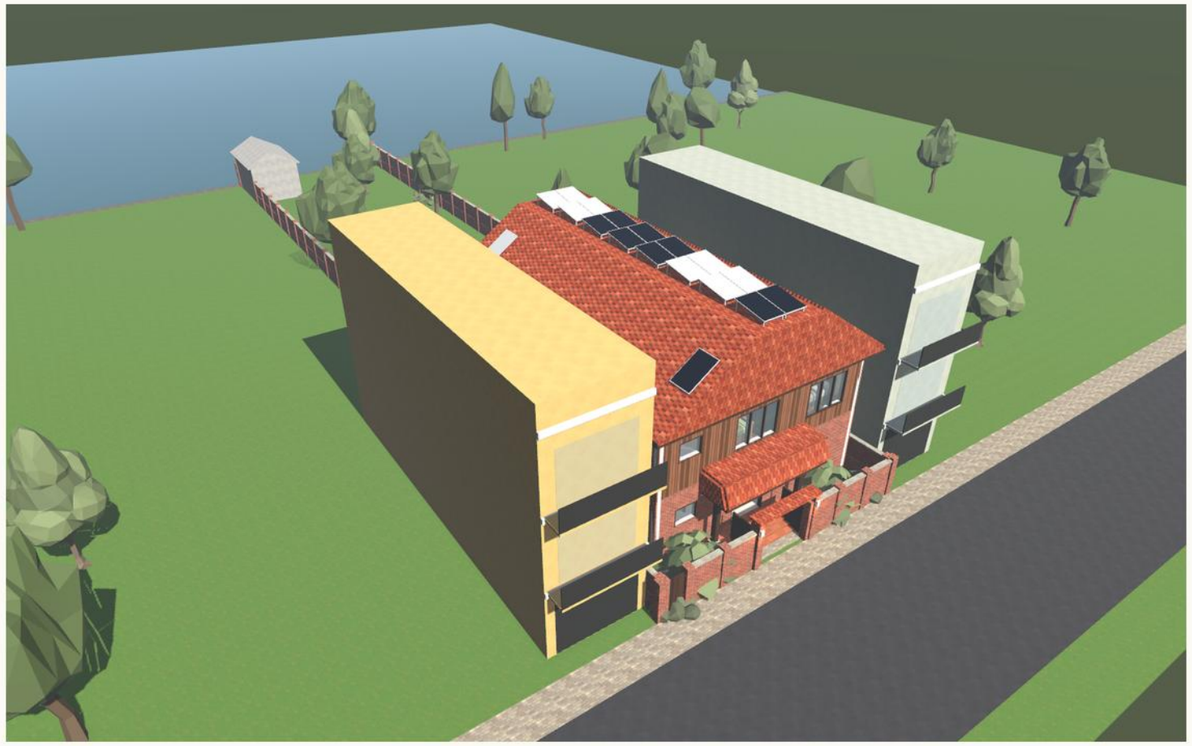


Sun study: winter solstice, 09:00 solar time

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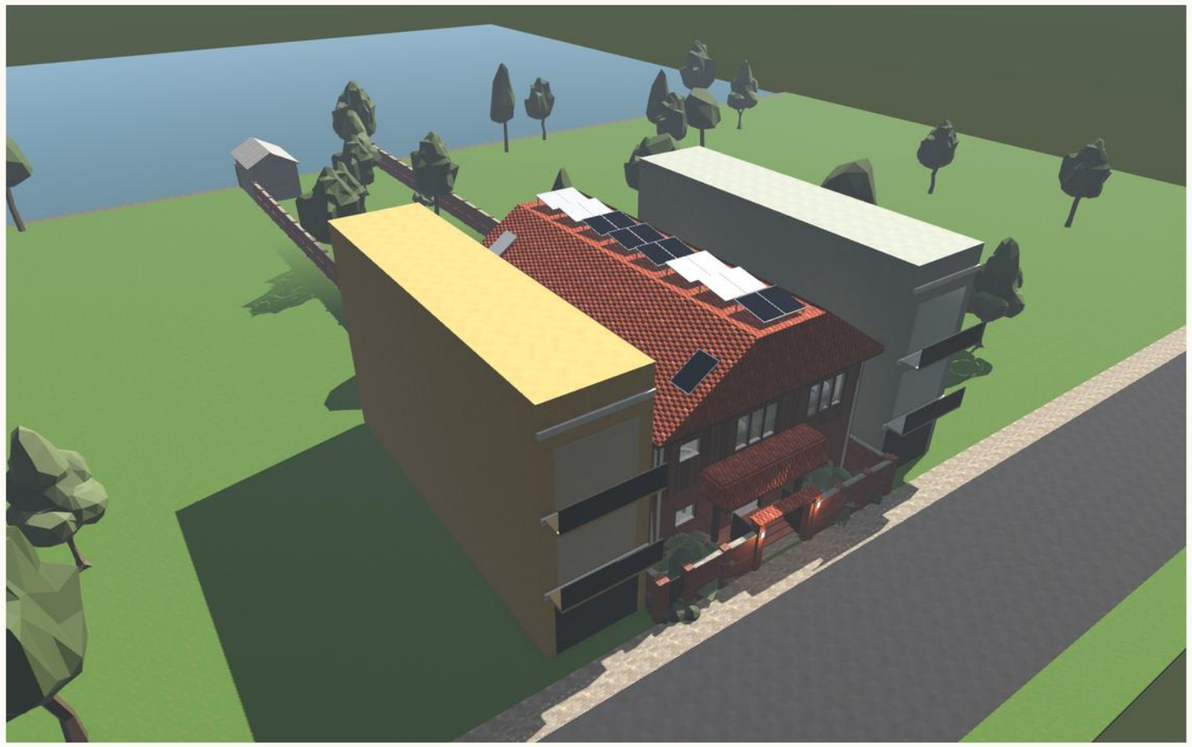
Interactive model: these are fixed viewpoints you can jump to with keys 1–9



Sun study: winter solstice, 12:00 solar time



Sun study: winter solstice, 15:00 solar time



Sun study: summer solstice, 09:00 solar time

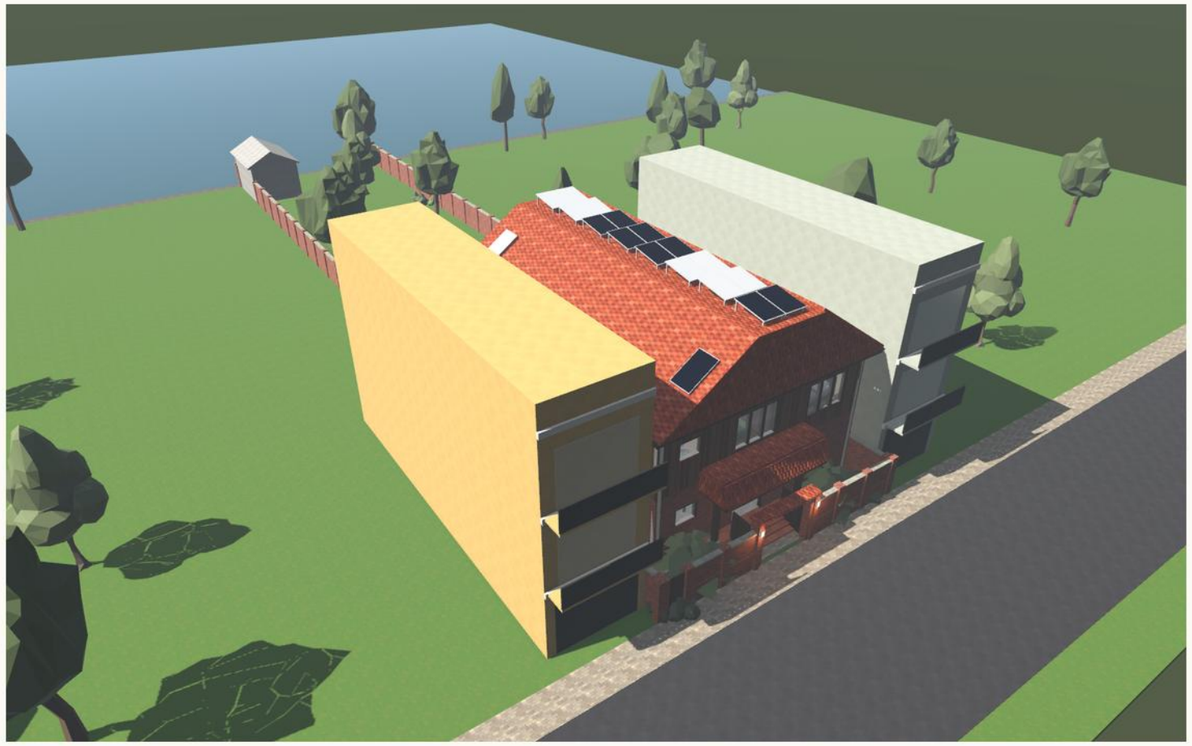


Sun study: summer solstice, 12:00 solar time

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# Walkthrough (Unity)

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Sun study: summer solstice, 15:00 solar time

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